

Mini-project 2

Design of a 2d truss crane

Project organization:

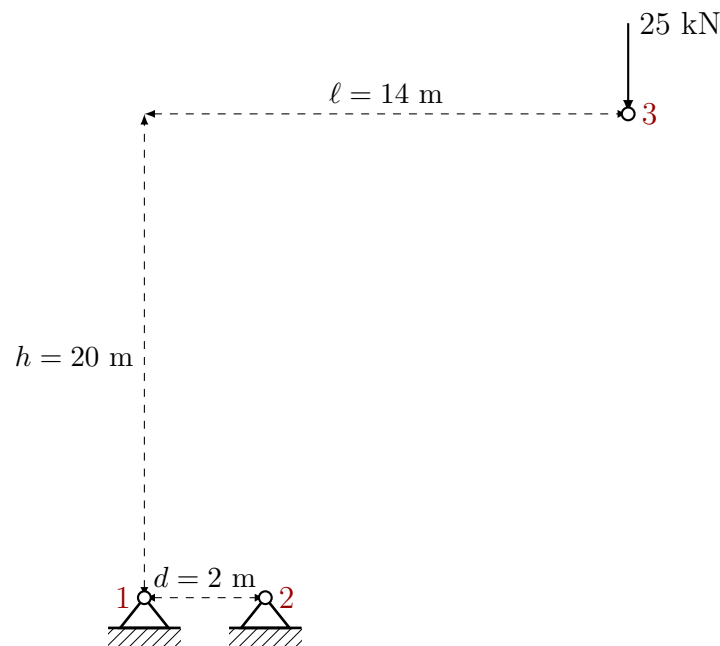
- Groups: 3 to 4 students
- 20% of final grade
- Pdf report: maximum 10 pages
- Programming language: **MATLAB**
- Submission: November 28, 2025
- Total workload: 40h (approx. 10h per student)

Your task is to design a crane truss structure for three different materials and make sure each individual structure you created fulfils the detailed static and dynamic requirements. The crane must lift a vertical point load of 25 kN applied at the tip of the jib (node 3), which extends vertically $h = 20$ meters above the ground level and horizontally $\ell = 14$ meters from the base left support (node 2). The two supports at the base (nodes 1 and 2) are fixed to the ground and are separated by a distance of $d = 2$ meters.

Anchor points and load configuration

The crane is fixed at the following supports on a horizontal plane:

- Base left: $(0, 0)$
- Base right: $(2, 0)$



Crane design requirements

The crane must be modeled as a planar truss. All joints are idealized as frictionless pinned connections. You are required to perform both a static and a dynamic analysis for each of the three materials listed below, treating each case as a distinct crane design. A distinct truss design must be developed for each of these three materials.

Material	E (GPa)	Density (kg/m ³)	Cross-Sectional area
Structural steel	200	7850	15–50 cm ²
Aluminum alloy	69	2700	20–70 cm ²
Engineered wood	12	600	30–100 cm ²

Static analysis

Each crane design must satisfy the following:

1. The structure must remain fully elastic with displacements below **60 mm**.
2. The total material volume must not exceed:
 - Structural steel: **1 m³**,
 - Aluminum alloy: **1.7 m³**,
 - Engineered wood: **2 m³**.

Dynamic analysis

A separate modal analysis must be performed for each of the three trusses designs corresponding to the different materials. The following requirement must be satisfied:

3. Compute the first three natural frequencies and mode shapes. The first natural frequency must *not* fall within **0 to 5 Hz**.

Bonus: To ensure dynamic stability and avoid resonance with common excitation sources, you should aim for a first natural frequency **above 20 Hz**, where feasible.

Deliverables

1. MATLAB:

- Nodes and elements definitions
- Stiffness and mass matrices assembly
- Static and modal analysis
- Plots and visualizations

2. Report:

- Crane layout with labeled pins and bars,
- Summary of MATLAB results (displacements, stresses, frequencies and mode shapes),
- Discussion of your findings.